



Question	Answer	Marks	Guidance
1	$2 + (12)(-2)x^{-3}$	B1	Correct differential but can be unsimplified.
	$2 + (12)(-2)(-2)^{-3} [= 5]$	*M1	Substitute $x = -2$ into <i>their</i> differential, which must contain x^{-3} .
	Either $(\text{their } 5) = \frac{y - (-1)}{x - (-2)}$ or $-1 = (\text{their } 5) \times (-2) + c \Rightarrow c =$	DM1	Attempt to find equation of tangent through $(-2, -1)$ with their numerical gradient obtained as described above.
	$y = 5x + 9$	A1	
		4	

Question	Answer	Marks	Guidance
2	$r = \frac{8 \sin^3 \theta}{4 \sin^2 \theta} [= 2 \sin \theta]$	B1	A correct unsimplified expression for r , this may only be seen in a correct S_{∞}
	$\frac{4 \sin^2 \theta}{1 - 2 \sin \theta} = \frac{1}{2}$	*M1	OE Correct S_{∞} with <i>their</i> expression for r .
	Or		
	$r = \frac{8 \sin^3 \theta}{4 \sin^2 \theta}$ and $\frac{4 \sin^2 \theta}{1 - r} = \frac{1}{2}$	B1	
	$1 - 8 \sin^2 \theta = \frac{8 \sin^3 \theta}{4 \sin^2 \theta}$	*M1	
	Then		
	Expect $8 \sin^2 \theta + 2 \sin \theta - 1 [= 0]$	DM1	OE and attempt to solve. Attempt to solve a three-term quadratic to find at least one value of $\sin \theta$. This can be implied by a final answer of $\theta = 0.253^{\circ}$ AWRT
	$[\theta =] \sin^{-1} \frac{1}{4}, 0.25$	A1	OE Clearly identify one answer only. Accept $\sin \theta = \frac{1}{4} \Rightarrow k = \frac{1}{4}$ etc.
		4	

Question	Answer	Marks	Guidance
3	$\left(\frac{1}{-1 \times 4}\right)a(4x-3)^{-1} + 2x$	B1	OE Do not accept $(-2+1)$ as equivalent to -1 .
	Apply correct limits, $x = 3$ & 1 , to <i>their</i> integral	*M1	<i>Their</i> integral must contain $(4x-3)^{-1}$. Condone using $x = 1$ and 3 .
	$\frac{-a}{36} + 6 - \left(\frac{-a}{4} + 2\right) = 12 \Rightarrow \left[\frac{8a}{36} + 4 = 12\right]$	DM1	OE Equate <i>their</i> linear unsimplified expression in a to 12 .
	$a = 36$	A1	
		4	

Question	Answer	Marks	Guidance
4(a)	$\{32[x^0]\}\{-120x\}\{+180x^2\}$	B1B1B1	Coefficients must be simplified. Ignore further terms in the expansion. SC B3: $2^5\left(1 - \frac{15}{4}x + \frac{45}{8}x^2 \dots\right)$ can be awarded full marks, ISW.
		3	
4(b)	$x = 0.01$ or $\frac{1}{100}$	B1	Identify 0.01 as the only value to substitute, condone any method.
	$(\text{Their } 32 - 120x + 180x^2)$ with $x = (\text{their } 0.01)$	M1	Clear use of <i>their</i> x -value from <i>their</i> solution of $2 - \frac{3}{2}x = 1.985$ and <i>their</i> answer to (a). Expect $32 - 120(0.01) + 180(0.01)^2 = 32 - 1.2 + 0.018$. Condone substitution into further terms in the expansion.
	30.8[18]	A1	Correct answer to at least 3sf coming from substitution into three or more terms. 'Correct' answers without appropriate working get A0.
		3	

Question	Answer	Marks	Guidance
5	Replace $\tan \theta$ with $\frac{\sin \theta}{\cos \theta}$	M1	
	Replace $\sin^2 \theta$ with $1 - \cos^2 \theta$ leading to a 3-term quadratic	*M1	To obtain a 3-term quadratic in $\cos \theta$. Accept a cubic that has a common factor of $\cos \theta$. Condone +/- sign errors during simplification.
	$9 \cos^2 \theta + \cos \theta - 4 [= 0]$	A1	OE
	Attempt to solve (<i>their</i> quadratic in $\cos \theta$) using a valid method	DM1	Only available for solution of a three-term quadratic. Correct values of $\cos \theta$ are 0.6134 and -0.7245 .
	Any two of ± 52.2 , ± 136.4	A1	AWRT as final answers. Any two correct answers between -180° and 180° as <i>their</i> final answers.
	All four values	A1	Condone other answers outside the range but no others between -180° and 180° . SC after M1 *M1 A1 DM0, correct answers can score B1 B1.
		6	

Question	Answer	Marks	Guidance
6	$a + 2(N - 1) = 55$	B1	SOI
	$\frac{1}{2} \times 3N(2a + 2(3N - 1)) = 5760$	B1	SOI
	Eliminate either a or N	M1	Using a valid substitution, expect either $a = 57 - 2N$ or $N = \frac{(57 - a)}{2}$. Condone +/- sign errors in <i>their</i> substitution or during simplification.
	$N^2 + 56N - 1920$ or $a^2 - 226a + 1953 [= 0]$	A1	All terms need not be collected on one side.
	$N = 24, a = 9$	A1 A1	Ignore presence of extra solutions for N and a .
		6	

Question	Answer	Marks	Guidance
7(a)	$3x^2 + 10x - 8 < 0$ or $3\left(x + \frac{5}{3}\right)^2 - \frac{49}{3} < 0$	*B1	OE Condone = or ≤ 0 , but use of $>$ or $\geq$ can only score B0B1B0. If there is no $<$, = or ≤ 0 , then a correct final answer implies < 0 here.
	-4 and $\frac{2}{3}$	B1	
	$-4 < x < \frac{2}{3}$	DB1 FT	OE Use of $\leq$ sign(s) gets DB0. Condone $x > -4$, $x < \frac{2}{3}$, $x > -4$ and $x < \frac{2}{3}$ but not $x > -4$ or $x < \frac{2}{3}$. FT on <i>their</i> critical values.
		3	
7(b)	Identify $x = -4$ as the x -value of the maximum point.	B1 FT	SOI FT on <i>their</i> lower critical value from part (a).
	$[y =] \frac{3}{3}x^3 + \frac{10}{2}x^2 - 8x [+c]$	B1	Correct integration of the three terms given. Condone $f(x) =$.
	Use of $y = 27$, $x = \text{their}\left(-4 \text{ or } \frac{2}{3}\right)$ in <i>their</i> integral.	M1	<i>Their</i> integral must be a cubic, <i>their</i> $\left(-4 \text{ or } \frac{2}{3}\right)$ must come from (a) or a restart.
	$c = -21 \Rightarrow y = x^3 + 5x^2 - 8x - 21$	A1	Do not ISW if they continue to find a straight line. Condone omission of final statement if y or $f(x) = x^3 + 5x^2 - 8x + c$ has been seen earlier.
		4	

Question	Answer	Marks	Guidance
8(a)	$\frac{1}{2}\pi - 2\left(\tan^{-1}\frac{4}{12} \text{ or } \sin^{-1}\frac{4}{\sqrt{160}} \text{ or } \cos^{-1}\frac{12}{\sqrt{160}}\right)$ Or $2\sin^{-1}\left[\frac{\frac{1}{2}EF}{AE} = \frac{\sqrt{32}}{\sqrt{160}} = \frac{\sqrt{2}}{\sqrt{10}}\right] \text{ or other valid method.}$	M1	Attempt a complete valid method for finding angle EAF. $\sqrt{160}$ may be replaced by 12.65 AWRT and $\sqrt{32}$ by 5.657 AWRT. Working in degrees should give $E\hat{A}F = 53.13^\circ$ which converts to the correct value in radians.
	$[E\hat{A}F = 0.927295\dots] = 0.9273$	A1	AG If decimals are used, at least one of 0.32175..., 0.6435 or 0.92729.. must be seen in the first method and 5.6569 and 12.649 or 0.46364 seen in the second method for the correct level of accuracy. Note: $E\hat{A}F = 53.14^\circ$ does not convert to the correct answer, so A0.
	Alternative Method for Question 8(a)		
	$12^2 - (24 + 24 + 32) = \frac{1}{2}(their AE \times their AF) \sin E\hat{A}F$ $\sin E\hat{A}F = \frac{2\{12^2 - (24 + 24 + 32)\}}{(their AE \times their AF)}$ $E\hat{A}F = \sin^{-1} \frac{2\{12^2 - (24 + 24 + 32)\}}{(their AE \times their AF)}$	M1	Attempt a complete valid method for finding angle EAF. Area of ABCD – (Area Δs ADF, ABE & CEF) = Area ΔAEF. $E\hat{A}F = \sin^{-1}\left(\frac{4}{5} \right).$ Using the cosine rule should lead to $\cos^{-1} \frac{3}{5}$.
	$\left[E\hat{A}F = \sin^{-1} \frac{4}{5} = \right] \frac{\frac{1}{2}EF}{AE}$	A1	AG
		2	

Question	Answer	Marks	Guidance
8(b)	$[AE \text{ or } AF =] \sqrt{160} \text{ or } \frac{4}{\sin \hat{EAB}}$	B1	OE Expect AWRT 12.65.
	$[r\theta =] (their\ 12.65) \times 0.9273$	M1	Use of $r\theta$ with $(their\ \sqrt{160})$. Note: using $r = 12$ scores M0.
	$[11.729... + 8 + 8 =] 27.7 \text{ AWRT}$	A1	
		3	
8(c)	$\frac{1}{2}(their\ 12.65)^2 \times 0.9273 [= 74.184]$	M1	Use $\frac{1}{2}r^2\theta$ for area of sector with $(their\ 12.65)$. Note: using $r = 12$ scores M0.
	$144 - 24 - 24 - \left\{ \frac{1}{2}(their\ 12.65)^2 \times 0.9273 \right\}$	M1	Attempt a complete method for finding area of shaded region. Condone use of $r = 12$.
	[Area =] 21.8	A1	AWRT
	Alternative Method for Question 8(c)		
	$\frac{1}{2}(their\ 12.65)^2 (0.9273 - \sin 0.9273) [= 10.183]$	M1	Area of the segment with $(their\ 12.65)$. Note: using $r = 12$ scores M0.
	$32 - \frac{1}{2}(their\ 12.65)^2 (0.9273 - \sin 0.9273)$	M1	Attempt a complete method for finding area of shaded region, condone use of $r = 12$.
	[Area =] 21.8	A1	AWRT
		3	

Question	Answer	Marks	Guidance
9(a)	Gradient of $PR = \frac{k-5}{2+13}$ or gradient of $RQ = \frac{k-1}{2-5}$	B1	Obtain at least one relevant gradient.
	$(\text{their gradient of } PR) \times (\text{their gradient of } RQ) = -1$ and attempt to simplify	M1	Clear of fractions and expand brackets. Expect $k^2 - 6k + 5 = 45$ OE. Condone +/-sign errors during simplification.
	$k^2 - 6k - 40 [= 0]$	A1	OE
	$k = 10, k = -4$	A1	
	Alternative Method for Question 9(a)		
	$PR^2 = (k-5)^2 + (2+13)^2$ or $RQ^2 = (k-1)^2 + (2-5)^2$	B1	Obtain at least one of PR^2 and RQ^2 . These expressions may be seen under a square root sign.
	$(\text{their } PR^2) + (\text{their } RQ^2) = 18^2 + 4^2$ and attempt to simplify	M1	Expect $2k^2 - 12k + 260 = 340$ OE. Condone +/-sign errors.
	$2k^2 - 12k - 80 [= 0]$	A1	OE
	$k = 10, k = -4$	A1	

Question	Answer	Marks	Guidance
9(a)	Alternative Method 2 for Question 9(a)		
	$x^2 + y^2 + 8x - 6y - 60 = 0$	B1	OE
	$2^2 + k^2 + 8 \times 2 - 6 \times k - 60 = 0$	M1	Substitution of $(2, k)$ into their circle equation.
	$k^2 - 6k - 40 [= 0]$	A1	OE
	$k = 10, k = -4$	A1	
		4	If none of the above marks are awarded, then SC B1 for using $k = 10$ to find the gradients of $QR (-3)$ and $PR (\frac{1}{3})$ and showing that their product is -1 (or other similar verification), and then stating that this shows that 10 is a possible value for k .

Question	Answer	Marks	Guidance
9(b)	[Centre is] $(-4, 3)$	B1	SOI
	Radius gradient = $\frac{(their\ 3) - 10}{(their\ -4) - 2} \left[= \frac{7}{6} \right]$	*M1	Attempt at finding the gradient of the radius using $(2, 10)$ and their centre, but not P, Q , the mid-point of PR or QR .
	Either $\frac{-1}{\left(\frac{their\ 7}{6} \right)} = \frac{y - 10}{x - 2}$ Or $10 = \frac{-1}{\left(\frac{their\ 7}{6} \right)} \times 2 + c \Rightarrow c = \dots$	DM1	OE Correct method to find the equation of the tangent using $\frac{-1}{their\ radius\ gradient}$ and $(2, 10)$.
	$y - 10 = -\frac{6}{7}(x - 2)$ or $y = \left(-\frac{6}{7} \right)x + \frac{82}{7}$	A1	OE Correct equation but not stated in the required form.
	$6x + 7y - 82 = 0$ or $82 - 6x - 7y = 0$	A1	All correct terms on one side, but condone them being in the wrong order.
	Alternative Method for Question 9(b)		
	$x^2 + y^2 + 8x - 6y - 60 = 0$	B1	OE Equation of the circle.
	$2x + 2y \frac{dy}{dx} + 8 - 6 \frac{dy}{dx} = 0$ Or $\frac{dy}{dx} = \frac{-2x - 8}{2} (85 - x^2 - 8x - 16)^{-\frac{1}{2}}$	*M1	Differentiate implicitly to arrive at an expression with two terms containing $\frac{dy}{dx}$. Or rearrange to make y the subject and differentiate to arrive at an expression of the form $f'(x) \times f(x)$.

Question	Answer	Marks	Guidance
9(b)	$2(2) + 2(10)\frac{dy}{dx} + 8 - 6\frac{dy}{dx} = 0$ $\text{Or } \frac{dy}{dx} = \frac{-2(2) - 8}{2} \left(85 - (2)^2 - 8(2) - 16 \right)^{-\frac{1}{2}}$	DM1	Substitute (2,10) into their implicit differential. Or substitute $x = 2$ into their expression for $\frac{dy}{dx}$.
	$y - 10 = -\frac{6}{7}(x - 2) \text{ or } y = -\frac{6}{7}x + \frac{82}{7}$	A1	OE Correct equation but not stated in the required form.
	$6x + 7y - 82 = 0$	A1	All correct terms on one side, but condone them being in the wrong order.
		5	

Question	Answer	Marks	Guidance
10(a)	$-9 \times 2(2x-5)^{-2} + 2$	B1	Correct differential.
	$(\text{their } -18(2x-5)^{-2} + 2) = 0$ and rearrange to form a quadratic. $\left[(2x-5)^2 = 9 \text{ or } 8x^2 - 40x + 32 = 0 \right]$	M1	Equating a two term $\frac{dy}{dx}$ to 0 and dealing correctly with the negative power. Their two term $\frac{dy}{dx}$ must contain $(2x-5)^{-2}$.
	$(1, -6)$ and $(4, 6)$	A1, A1	A1 for two correct x -values or one correct point, second A1 for all correct.
		4	
10(b)	$-18 \times -2 \times 2(2x-5)^{-3} \left[= 72(2x-5)^{-3} \text{ or } \frac{144x-360}{(2x-5)^4} \right]$	B1 FT	Following through on <i>their</i> first derivative which must contain $(2x-5)^{-2}$.
	Use $(\text{their } x=1 \text{ and } x=4)$ in $(\text{their } 72(2x-5)^{-3})$ To determine the nature of both turning points.	M1	Substitute x -coordinate of each stationary point and determine their nature. Nature of the turning points must correctly follow from <i>their</i> values of $\frac{d^2y}{dx^2}$.
	For $x=1$, $\left[\frac{d^2y}{dx^2} = \right] -\frac{72}{27}$ or $< 0 \Rightarrow$ maximum For $x=4$, $\left[\frac{d^2y}{dx^2} = \right] \frac{72}{27}$ or $> 0 \Rightarrow$ minimum	A1	CWO
		3	

Question	Answer	Marks	Guidance
10(c)(i)	$(1, -13)$	B1	
		1	
10(c)(ii)	$[y =] \frac{9}{2(x \pm 3) - 5} + 2(x \pm 3) - 5 \pm 7$	M1	Application of $\begin{pmatrix} -3 \\ 7 \end{pmatrix}$ to the original expression for C but condone $+/-$ sign errors.
	$[y =] - \left(\frac{9}{2(x \pm 3) - 5} + 2(x \pm 3) - 5 \pm 7 \right)$	M1	SC B1 for $[y =] - \left(\frac{9}{2x - 5} + 2x - 5 \right)$.
	$y = -\frac{9}{2x + 1} - 2x - 8$	A1	Answer must be in this format; the ' $y =$ ' can be implied by earlier inclusion.
		3	

Question	Answer	Marks	Guidance
11(a)	$[f(x) =](x + 2a)^2 - 4a^2 + a$	B1	
	$(their(-4a^2 + a)) = -33$	M1	Condone $\geq$ or $>$.
	$4a^2 - a - 33 [= 0]$	B1	Condone $\leq$ or $<$.
	$-\frac{11}{4}, 3$	A1	OE Do not ISW if their final answer is given as a range or if one of the answers is rejected.
	Alternative Method for Question 11(a)		
	$x = -2a, y = -4a^2 + a$	B1	The co-ordinates of the minimum point (likely to be either from differentiation or completing the square).
	$(their - 4a^2 + a) = -33$	M1	Their y-coordinate equated to -33 . Condone $\geq$ or $>$.
	$-4a^2 + a + 33 [= 0]$	B1	OE Condone $\geq$ or $>$.
	$-\frac{11}{4}, 3$	A1	OE Do not ISW if their final answer is given as a range or if one of the answers is rejected.

Question	Answer	Marks	Guidance
11(a)	Alternative Method 2 for Question 11(a)		
	$x^2 + 4ax + a + 33 [=0]$	B1	Condone $\geq$ or $>$.
	$(4a)^2 - 4[1](a + 33) [=0]$	M1	Condone $\geq$ or $>$.
	$16a^2 - 4a - 132 [=0]$	B1	OE Condone $\geq$ or $>$. Accept $4a^2 - a - 33$ or multiples thereof.
	$-\frac{11}{4}, 3$	A1	OE Do not ISW if their final answer is given as a range or if one of the answers is rejected.
		4	

Question	Answer	Marks	Guidance
11(b)	$[g(x) =] \frac{1}{2}(x^3 + 4)$	B1	Expression for $g(x)$. SOI
	Either		
	$[g(0) =] \frac{1}{2}((0)^3 \pm 4) \quad [g(0) = 2]$	M1	Replacing x with 0 in <i>their</i> $g(x)$.
	$gg(0) = 6$	A1	
	$f(\text{their } 6) [=36 + 24a + a]$	M1	
	Or		
	$fg(x) = \left(\frac{x^3 + 4}{2}\right)^2 + 4a\left(\frac{x^3 + 4}{2}\right) + a$ or $gg(x) = \frac{\left(\frac{x^3 + 4}{2}\right)^3 + 4}{2} \quad \left[= \frac{(x^3 + 4)^3}{16} + 2 \right]$	M1	Either composite function using <i>their</i> $g(x)$.
	$fgg(x) = \left\{ \frac{(x^3 + 4)^3}{16} + 2 \right\}^2 + 4a \left\{ \frac{(x^3 + 4)^3}{16} + 2 \right\} + a$	A1	Complete algebraic expression for $fgg(x)$.
	$fgg(0) = \left\{ \frac{(0 + 4)^3}{16} + 2 \right\}^2 + 4a \left\{ \frac{(0 + 4)^3}{16} + 2 \right\} + a$	M1	Replacing x with 0 in <i>their</i> $fgg(x)$.

Question	Answer	Marks	Guidance
11(b)	Then		
	$36 + 24a + a = 96$	A1	OE
	$[a =] \frac{12}{5}$	A1	OE
		6	